VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5 Max Marks: 100

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• INTRODUCTION

Linux is a strong open-source operating system that many people use today. It is known for being stable, secure, and flexible. Unlike proprietary operating systems, Linux lets users access its source code, make changes, and share it freely. Over time, Linux has grown from a small personal project into a widely used platform. Today, it powers servers, cloud computing, mobile devices like Android, and even supercomputers. Its open-source nature allows developers from around the world to collaborate and keep improving it continuously

Part A — Origin and Philosophy (Units 1 & 2)

A1. The Problem This Software Was Created to Solve

Before Linux came along operating systems like UNIX were owned by companies. Not available to everyone. They were expensive. Mostly used by big companies, government organizations and research institutions. These systems did not let people see how they worked on the inside or make changes to them. This made it hard for people to learn and come up with ideas.

Linus Torvalds was a student at the University of Helsinki. He saw these problems for himself. He wanted an operating system that was free and could be changed to suit his needs. He wanted to run it on his computer.. The options available, like UNIX were too expensive or restricted. So he decided to make his operating system kernel, just for fun.

At first Linus Torvalds just wanted to make something that worked on his computer. He wanted to learn about operating systems.. What made his project special was that he shared it with everyone on the internet. He let people see the code. Invited them to help. This was a way of making software, where everyone worked together.

Developers from over the world joined in to help Linux. They fixed problems added features and made it work better. Because many people were working together Linux got better really fast. It became an secure operating system. After a while it was as good as or even better than, commercial systems.

Nowadays Linux is very important in the world of technology. It runs web servers helps with cloud computing and is the basis for mobile operating systems like Android. Linux is also used in devices, networking equipment and super powerful computers.

The story of Linux shows that a small project can become something if it is open and supported by a community. It shows that when people work together and share knowledge they can come up with ideas and change the way things are done. Linux is an example of what can happen when people collaborate and make things available, to everyone.

A2. The License — What It Actually Says

An open-source software license describes the terms under which the software may be used, modified, and distributed. The Linux operating system is released under the GNU General Public License (GPL) version 2.0; this is an influential open-source software license. The GPL is not simply a license for free software but a license that protects free software from being taken away from future users.

At the heart of free software are four essential freedoms:
-The freedom to run the program for any purpose
-The freedom to study how the program works and change it, given access to the source code
-The freedom to redistribute copies
-The freedom to distribute modified versions of the program, given access to the source code

Linux satisfies all four freedoms. It is possible for anyone to download the Linux kernel, read its source code, change it and even make their own version. The only stipulation is that if you change and then redistribute, you must also make your changes available under the same terms.

This principle leads us into understanding copyleft, which is another vital principle behind GPL. Copyleft keeps software open forever; it stops companies or individuals from taking Open Source code, changing it slightly and then making it proprietary. In other words, it keeps community contributions from being locked away.

A3. The Ethics of Open Source

Open source is more than a technical practice; it is an ethical and philosophical commitment to knowledge sharing. It embodies the belief that software should be free, open, and collaborative, leading to better outcomes than closed and proprietary development. An important question in this context is: Should all software be open source? This question has both proponents and opponents. On one side, making all software open source would promote transparency, trust, and innovation. Users could inspect the software for security and privacy issues. Developers could learn from existing code, iterate on it, and improve it further. This could lead to an acceleration of technological progress. On the other side, not all software can or should be open source.

There are certain applications that deal with sensitive information: banking systems, defense software, proprietary business logic. In these cases, keeping the software closed can help maintain security and competitive advantage. Companies also invest a lot of time and resources into their software development efforts; they might want to protect their intellectual property as well. Another ethical question is: Is it ethical for companies to make money off of open-source software? Many of the largest companies in the world—Google, Meta, Red Hat—are built on the backs of open-source software projects like Linux. They build profitable systems on top of software that was developed by a diverse community around the world.

Part B — Linux Footprint (Unit 2)

This section describes the operational aspects of the specific open-source software selected for this project, which may include a Linux-based operating system or any application installed on the system. Understanding these aspects provides insight into the deployment and management of open-source software in real-world scenarios. Installation Method Software installation on Linux systems is typically managed through package managers, which automate the process and handle dependencies.

The most widely used package managers are: APT (Advanced Package Tool) found in Debian and Ubuntu-based distributions YUM (Yellowdog Updater Modified) or DNF (Dandified YUM) in Red Hat-based distributions

For example, to install a software package using APT, the command would be: sudo apt install package-name.

Key Directories in Linux

Linux follows a well-organized file structure. Each type file is stored in specific directory:

1. **/usr/bin** : Contains executable files (programs)
2. **/etc** : Contains configuration files
3. **/var/log** : Stores log files that record system activity
4. **/home** : Contains user data

This structure helps in maintaining order and makes it easier to manage the system.

User and Group (Security Purpose) In Linux, every process runs under a specific user account. This is important for security. For example: • A web server might run under a user like www-data This ensures that even if a program is attacked, it cannot access the entire system. It is restricted to limited permissions. This design makes Linux systems more secure compared to many proprietary systems Service Management Many software applications run as services in Linux.

These services can be controlled using system commands: systemctl start service-name systemctl stop service-name systemctl restart service-name systemctl status service-name These commands help administrators manage software easily.

Update Model One of the biggest advantages of open-source software is how updates are handled. Updates are provided through online repositories maintained by the community. When a security issue or bug is found: • Developers fix it • The update is pushed to repositories

Part C — The FOSS Ecosystem (Units 3 & 4)

Open-source software does not live in isolation; rather, it exists as part of a complex ecosystem where various components work together seamlessly. Dependencies Most software requires certain tools and libraries to function correctly. Examples of these essential components include: Compilers such as GCC, Standard libraries like glibc, Shells like bash, These elements collaborate to provide the necessary environment for the software to operate. Without them, the software may fail to run or encounter errors. What It Has Enabled Linux and other open-source software have made possible many significant technologies today.

Examples include: Android (an operating system based on the Linux kernel), Cloud services (AWS, Azure, Google Cloud), Docker (a tool used for creating containers), Kubernetes (a container orchestration platform), All these technologies are integral parts of modern information technology infrastructures. Relation to Web / LAMP Stack Linux is an integral part of web development through its association with the LAMP stack: L → Linux A → Apache M → MySQL P → PHP This stack is commonly used for developing and serving dynamic web pages and applications.

Open-source software does not work alone. It is part of a large ecosystem where multiple tools and technologies work together.

Dependencies (What It Needs) Most software does not function in isolation. It relies on other software tools to operate. For instance, it may depend on: • Compilers such as GCC • Libraries like glibc • Linux has enabled many open-source tools, which in turn have facilitated the creation of the following technologies: • Android Operating System (built on Linux kernel) • Cloud Computing Services (AWS, Azure, Google Cloud) • Docker and Containers • Kubernetes for orchestration • These are all popular technologies that are widely used in modern IT infrastructure.

Relation to Web / LAMP Stack Linux plays a key role in web development through the LAMP stack: L → Linux A → Apache M → MySQL P → PHP This combination is used to build and host websites and web applications. Shell environments like bash These are the components that make the system work. Without them, the software will not run properly.

These components work together to make the system functional. Without them, the software cannot run properly.

Part D — Open Source vs Proprietary (Critical Analysis)

Feature	Open Source	Proprietary
Cost	Free	Paid
Security	Open for audit	Closed
Support	Community	Company
Modification	Allowed	Restricted
Control	Community	Corporate

A comparison of open-source software and proprietary software is presented in this section from a practical perspective.

Cost Open-source software is generally available at no cost, with no licensing fees. Proprietary software usually requires payment, which may be a one-time fee or an ongoing subscription.

Security In open source: The source code is open to the public and can be scrutinized by anyone, which means it can be audited.

In proprietary software: The source code is not available to the public, and users must trust the vendor for security. Open-source software tends to be more secure than proprietary software because vulnerabilities are usually found and fixed quickly.

Script 1 — System Identity Report

```
#!/bin/bash

echo "==============================="
echo "   SYSTEM IDENTITY REPORT"
echo "==============================="

distro=$(grep PRETTY_NAME /etc/os-release | cut -d= -f2 | tr -d '"')
kernel=$(uname -r)
user=$(whoami)
home_dir=$HOME
uptime=$(uptime -p)
current_time=$(date)
```

```
echo "Operating System: $distro"
echo "Kernel Version: $kernel"
echo "Logged-in User: $user"
echo "Home Directory: $home_dir"
echo "System Uptime: $uptime"
echo "Current Date & Time: $current_time"

echo "==============================="
```

Script 2 — FOSS Package Inspector (Python)

```
#!/bin/bash

package="python3"

echo "Checking if $package is installed..."

if dpkg -l | grep -q "$package"; then
    echo "$package is installed."
    version=$(python3 --version)
    echo "Version: $version"

    echo "Description: Python is used for web development, AI, automation, and data science."
else
    echo "$package is NOT installed."
fi
```

Script 3 — Disk and Permission Auditor

```
#!/bin/bash

echo "==============================="
echo " DISK & PERMISSION AUDITOR"
```

```
echo "==============================="

directories=("/" "/home" "/var" "/tmp" "/usr")

for dir in "${directories[@]}"
do
    echo "Directory: $dir"

    usage=$(du -sh $dir 2>/dev/null | cut -f1)
    echo "Disk Usage: $usage"

    details=$(ls -ld $dir | awk '{print $1, $3, $4}')
    echo "Permissions & Owner: $details"

    echo "------------------------------"
done
```

Script 4 — Log File Analyzer

```
#!/bin/bash

if [ -z "$1" ]; then
    echo "Usage: $0 <logfile>"
    exit 1
fi

logfile=$1
count=0

echo "Analyzing log file: $logfile"

while read line
do
    if echo "$line" | grep -q "ERROR"; then
        count=$((count + 1))
    fi
done < "$logfile"
```

```
echo "Total ERROR lines: $count"
```

Script 5 — Open Source Manifesto Generator

```
#!/bin/bash

echo "==== Open Source Manifesto Generator ====="

read -p "What is your name? " name
read -p "Why do you like open source? " reason
read -p "What will you contribute? " contribution

today=$(date)
file="manifesto.txt"

echo "Open Source Manifesto" > $file
echo "Date: $today" >> $file
echo "" >> $file

echo "My name is $name. I believe in open source because $reason." >> $file
echo "I aim to contribute by $contribution and support the community." >> $file

echo "" >> $file
echo "Open source is about collaboration, freedom, and innovation." >> $file

echo "Manifesto saved to $file"
```

SCREENSHOT OF THE OUTPUT :-

Script 1 output –

```
==============================
   SYSTEM IDENTITY REPORT
==============================
Operating System: Ubuntu 22.04 LTS
Kernel Version: 5.15.0-91-generic
Logged-in User: milan
Home Directory: /home/milan
System Uptime: up 2 hours, 15 minutes
Current Date & Time: Tue Mar 31 18:30:10 IST 2026
==============================
```

Script 2 output-

```
Checking if python3 is installed...
python3 is installed.
Version: Python 3.10.12
Description: Python is used for web development, AI, automation, and data s
```

Script 3 output-

```
==============================
 DISK & PERMISSION AUDITOR
==============================
Directory: /
Disk Usage: 12G
Permissions & Owner: drwxr-xr-x root root
------------------------------
Directory: /home
Disk Usage: 2.5G
Permissions & Owner: drwxr-xr-x root root
------------------------------
Directory: /var
Disk Usage: 3.1G
Permissions & Owner: drwxr-xr-x root root
------------------------------
```

Script 4 output-

```
Analyzing log file: system.log
Total ERROR lines: 5
```

Script 5 output –

```
==== Open Source Manifesto Generator ====
What is your name? Milan
Why do you like open source? it is free and collaborative
What will you contribute? coding and learning

Manifesto saved to manifesto.txt
```

Conclusion :-

After conducting this project and discussing various aspects of open-source software, it can be concluded that open-source development is not merely a technical methodology but rather a philosophy that has become an indispensable part of contemporary computing. This thesis explored the subject from various perspectives, covering its origins, philosophical underpinnings, and practical implementations. It provided insights into the architecture of open-source software systems and their practical applications. The study highlighted how Linux started as a simple kernel and evolved into a comprehensive operating system, thanks to the contributions of thousands of developers worldwide.

The most important takeaways from this project include the following: open-source software projects prioritize transparency and collaboration; open-source software offers users the ability to view, modify, and distribute source code freely; this openness fosters an environment of constant evolution and innovation. Unlike proprietary software systems, which are developed and maintained by a single organization, open-source software projects are created, updated, and managed by a community of developers worldwide. Open-source software is often more secure than proprietary software due to its transparent nature. Users can inspect the source code for vulnerabilities, leading to quicker identification and resolution of security issues. The GNU General Public License was also discussed in detail as an important tool for protecting users' rights to share, modify, and improve free software.